Healing process

Learning outcome

- Recognize growth factors involving in tissue repair and healing
- Identify steps involving in tissue healing and repair
- State the complication following healing

Growth factor

- Healing involves an orderly sequence of events which includes regeneration and migration of specialized cells, angiogenesis, proliferation of fibroblasts and related cells, matrix protein synthesis and finally cessation of these processes.
- These processes are mediated by **growth factors**.

Growth Factors

- Very important in tissue repair.
- Actions:
  - stimulate cell division and proliferation
  - promote cell survival
- Huge list! Usually have “GF” in name:
  - EGF
  - TGF
  - PDGF

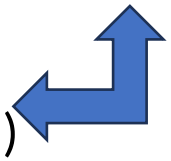
Sources of Growth Factors:

- Following injury, growth factors may be derived from a number of sources such as:

1. Platelets, activated after endothelial damage,
2. Damaged epithelial cells,
3. Circulating serum growth factors,
4. Macrophages, or
5. Lymphocytes recruited to the area of injury

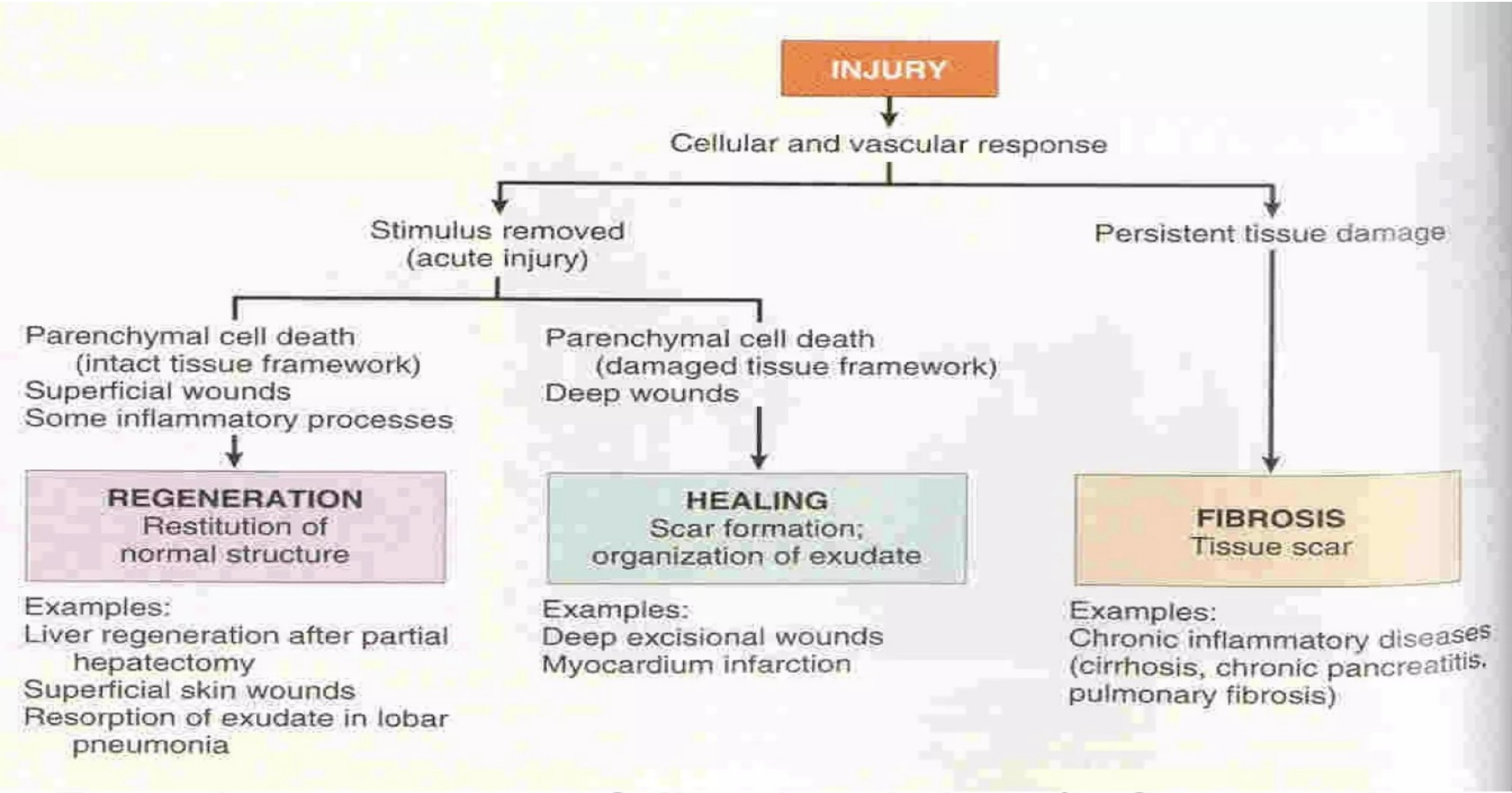
Growth factors

- VEGF----Angiogenesis (migration & proliferation of endothelial cells)
- TGF-B (supress endothelial proliferation).
- PDGF (Recruitment of smooth muscle cell)



Stabilize newly formed blood vessels

- ECM ---- (Angiogenesis)
- MMPs---- (ECM enzymes) permit remodeling & extension of vascular tube



Repair responses following injury/inflammation

Following tissue injury, sequential phases for healing

- 1- Starting with a **hemostatic plug** composed of platelets to stop the bleeding and provide a scaffold for fibrin deposition.
- 2- Then after **inflammation** occurs and resolves,
- 3- various cell types (epithelial cells, endothelial cells and pericytes, fibroblasts) **proliferate** and migrate to close the now clean wound, initially via formation of **granulation tissue** (loose connective tissue with new thin-walled delicate capillaries and often admixed inflammatory cells) and
- 4- subsequent replacement by increasing amounts of collagen to form a **fibrous scar (maturation phase)**

Replacement of necrotic tissue

- resorption by macrophages
- dissolution by enzymes
- replacement by granulation tissue
  - uniform mechanism irrespective of inicial trigger
  - the same microscopic appearance

  - angiogenesis
  - migration and proliferation of fibroblasts
  - deposition of ECM
  - maturation and reorganization

Repair by Connective Tissue Deposition

- **If repair cannot be accomplished by regeneration alone**
  - **it occurs by replacement of the injured cells with connective tissue,**
    - **leading to the formation of a scar,**
    - **or by a combination of regeneration of some residual cells and scar formation.**

❖Scarring;

Deposition of connective tissue (fibrosis), when the **tissues are incapable of regeneration or supporting structure is too damaged to support regeneration alone**.

❖Purpose;

is to provide enough structural stability for the injured tissue to function, not restoration

❖ This process is also known as organization when the fibrosis occurs in a tissue space occupied by inflammatory exudate (e.g. organizing pneumonia in the lung)

❖**Fibrosis in parenchymal organs :** responsible for organ dysfunction / failure

- End stage kidney dis.
- e.g. liver cirrhosis,
- fibrosing diseases of the lung

❖Mechanism of fibrosis (excessive deposition of collagen and other ECM components in tissue) is basically similar to scar formation in the skin during tissue repair.

Steps in Scar Formation

- Repair by connective tissue deposition consists of sequential processes that follow tissue injury and the inflammatory response.
- **Angiogenesis** is the formation of new blood vessels, which supply nutrients and oxygen needed to support the repair process.
- Newly formed vessels are leaky because of incomplete interendothelial junctions and because VEGF, the growth factor that drives angiogenesis, increases vascular permeability.
- This leakiness accounts in part for the edema that may persist in healing wounds long after the acute inflammatory response has resolved.
- **Formation of granulation tissue.** Migration and proliferation of fibroblasts and deposition of loose connective tissue, together with the vessels and interspersed leukocytes, form *granulation tissue*.
- The **term *granulation tissue*** derives from its pink, soft, granular gross appearance, such as that seen beneath the scab of a skin wound.
- Its **histologic appearance** is characterized by proliferation of fibroblasts and new thin-walled, delicate capillaries (angiogenesis), in a loose extracellular matrix, often with admixed inflammatory cells, mainly macrophages. Granulation tissue progressively invades the site of injury; the amount of granulation tissue that is formed depends on the size of the tissue deficit created by the

- **Remodeling of connective tissue.** Maturation and reorganization of the connective tissue (remodeling) produce the stable fibrous *scar*. The amount of connective tissue increases in the granulation tissue, eventually resulting in the formation of a scar, which may remodel over time.

Angiogenesis:

- Process of new blood vessel formation from existing vessels.
- Seen in healing, as well as development of collateral blood supply in ischemia and tumor growth.

•Hallmark of healing (tissue of repair)

Granulation tissue

pink soft granular appearance

richly vascularized

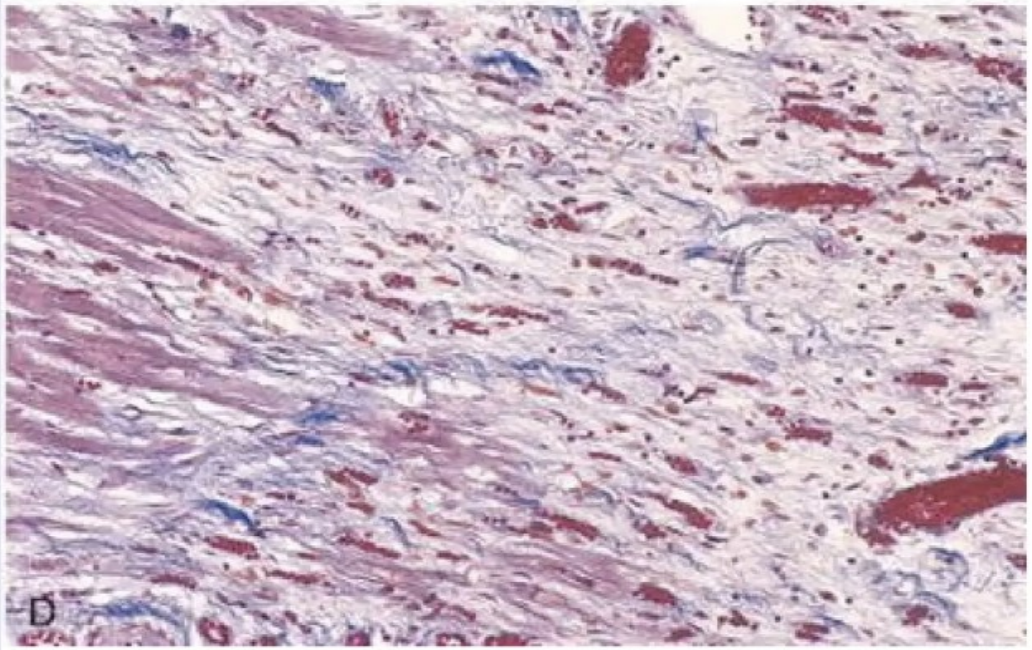
highly cellular

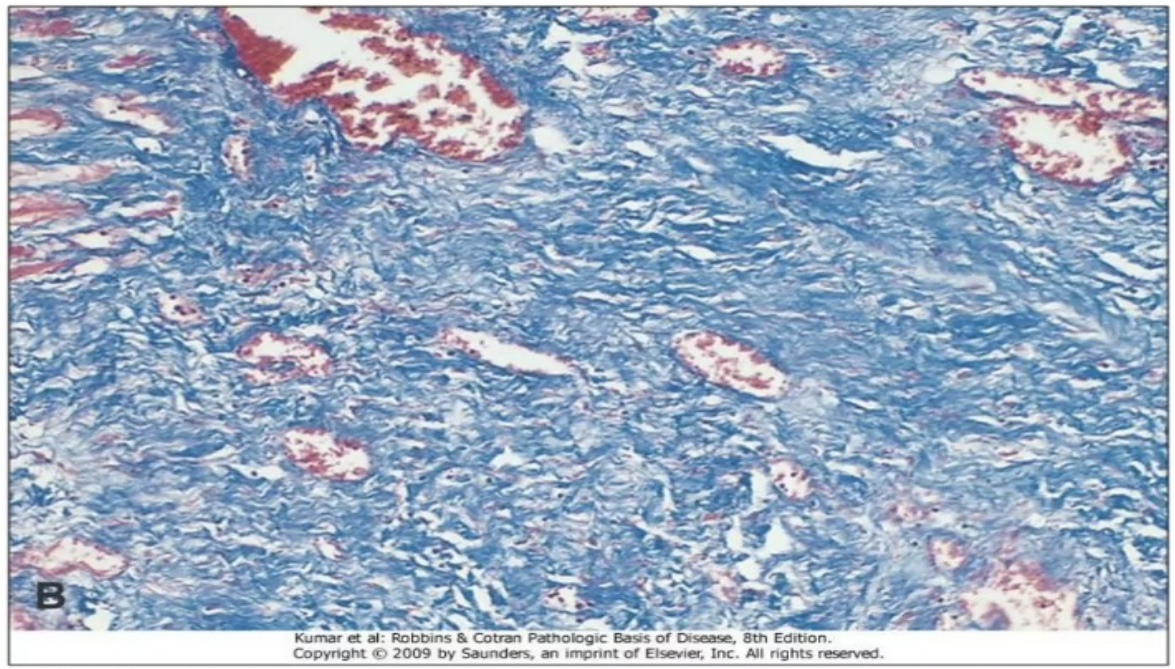
myxoid matrix

inflammatory cells

e.g. surface of wounds, bottom of ulcers

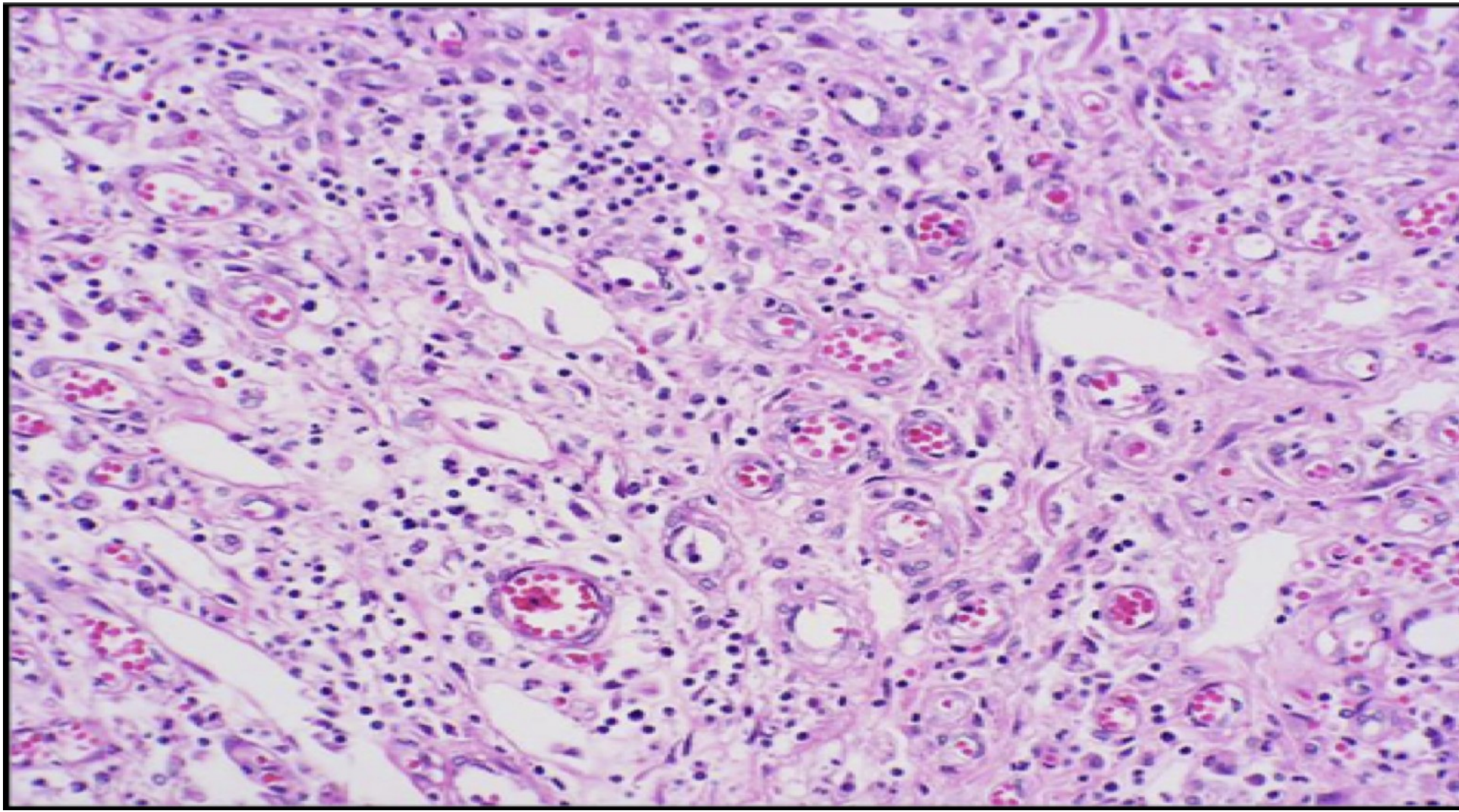






Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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Granulation tissue showing numerous blood vessels, edema, and a loose extracellular matrix containing occasional inflammatory cells. Collagen is stained blue by the trichrome stain; minimal mature collagen can be seen at this point. **B,** Trichrome stain of mature scar, showing dense collagen, with only scattered vascular channels.



Selected Clinical Examples of Tissue Repair and Fibrosis

- two clinically significant types of repair—the healing of skin wounds (cutaneous wound healing) and
- fibrosis in injured parenchymal organs.

Healing of Skin Wounds

- Process that involves both epithelial regeneration and the formation of connective tissue scar and is thus illustrative of the general principles that apply to healing in all tissues.
- Based on the nature and size of the wound, the healing of skin wounds is said to occur by first or second intention.

Healing by first intention (primary union):

clean surgical wound

- When the injury is relatively limited (with closely apposed wound edges), involving mostly the epithelial layer, with only focal disruption of the epithelial basement membrane and death of relatively few epithelial and connective tissue cells e.g. a clean uninfected surgical incision



first intention healing

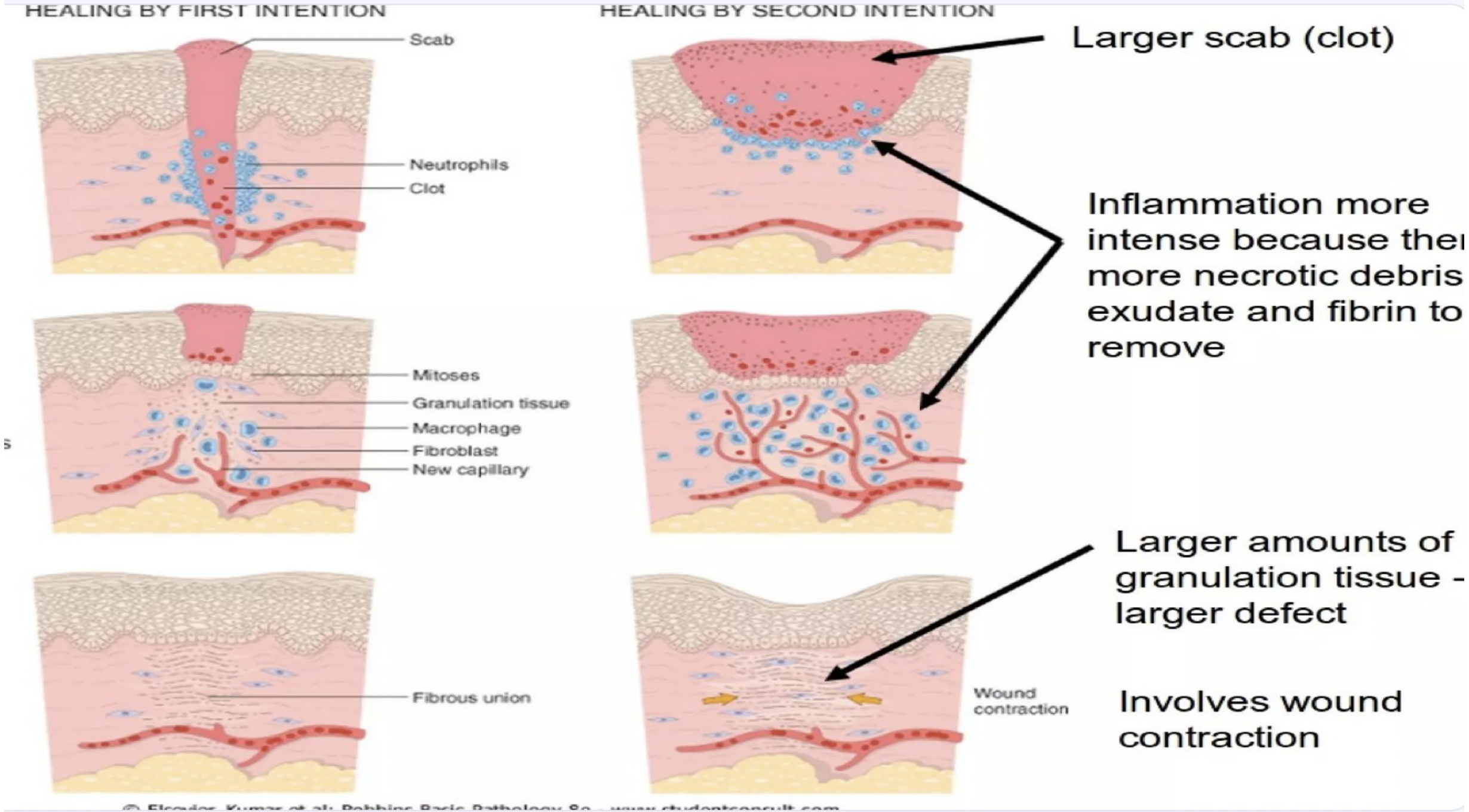
Healing by second intention (secondary union):

- When the wound is larger and gaping with edges not apposed.
- This means the fibrin clot is therefor also larger, with more exudate, necrosis and inflammation that can cause greater secondary inflammation-mediated injury.
- The larger amount of granulation tissue formed also results in a greater mass of scar tissue.
- Wound contraction by the action of myofibroblasts is important to help close the wound by decreasing the wound surface area

Healing by second intention (secondary union):



second intention healing


HEALING BY FIRST INTENTION
HEALING BY SECOND INTENTION
Scab
Neutrophils
Clot
Mitoses
Granulation tissue
Macrophage
Fibroblast
New capillary
Fibrous union
Wound contraction
Larger scab (clot)
Inflammation more intense because ther more necrotic debris exudate and fibrin to remove
Larger amounts of granulation tissue - larger defect
Involves wound contraction

Wound Healing Mechanism

- Cutaneous wound healing commonly can be divided into three phases
  1. Inflammatory phase
  2. Proliferative phase
  3. Remodeling / maturational phase

- ○ After the wound occurs, coagulation pathways are activated, forming a blood clot that stops bleeding and supports migrating cells attracted by cytokines, chemokines and growth factors
- ○ Within 24 hours, neutrophils are seen at the incision margin and migrating towards the fibrin clot. Within 24-48 hours. epithelial cells from both skin edges migrate and proliferate along the dermis, forming a thin and continuous epithelial layer that closes the wound
- ○ By day 3, neutrophils are replaced by macrophages, and blood clot replaced by granulation tissue, which peaks in day 5 to 1 wk

- Fibroblasts progressively migrate into the edematous vascular granulation tissue to lay down collagen
- ○ During the 2nd week, there is conversion of the granulation tissue into a collagenous scar, which is eventually covered an essentially normal epidermis by the end of the first month
- ○ The tensile strength of the wound increases over the first 2 months of healing by cross- linking of collagen fibers and increased fiber size, and reaches a maximum of 70-80% of normal by 3 months

❖Complication and abnormalities of (wound and healing process)

Abnormalities in tissue repair

1-Deficient scar formation (chronic wounds) & ulcer: Due to local and systemic factors affecting tissue repair. It could be due to;

- **Venous leg ulcers** (due to venous congestion causing poor oxygenation of local tissues),
- **Arterial ulcers** (in patients with peripheral vascular disease),
- **Diabetic ulcers**, pressure sores (due to prolonged mechanical tissue compression and local ischemia)

- **Local defect like wound dehiscence** (when a wound reopens)
  - e.g. due to
    - Obesity
    - Malnutrition
    - Infections



2- Contracture: It prone to develop on the palms, soles, the anterior aspect of the thorax. Excess contraction of the wound, resulting in **deformities** and impairment of **joint** movement. Commonly after serious burns



3-Excessive scarring: Can cause

Hypertrophic scar is raised, erythematous, and remains within the boundaries of the original trauma or wound.

A keloid, on the other hand, extends beyond the confines of the wound and is more nodular.

A-Hypertrophic scars (raised scar that regresses over months, generally after thermal or traumatic injuries involving the deep dermis) &



B-keloids (when the scar tissue grows beyond the boundaries of the original wound and does not regress; seems to be an individual predisposition).

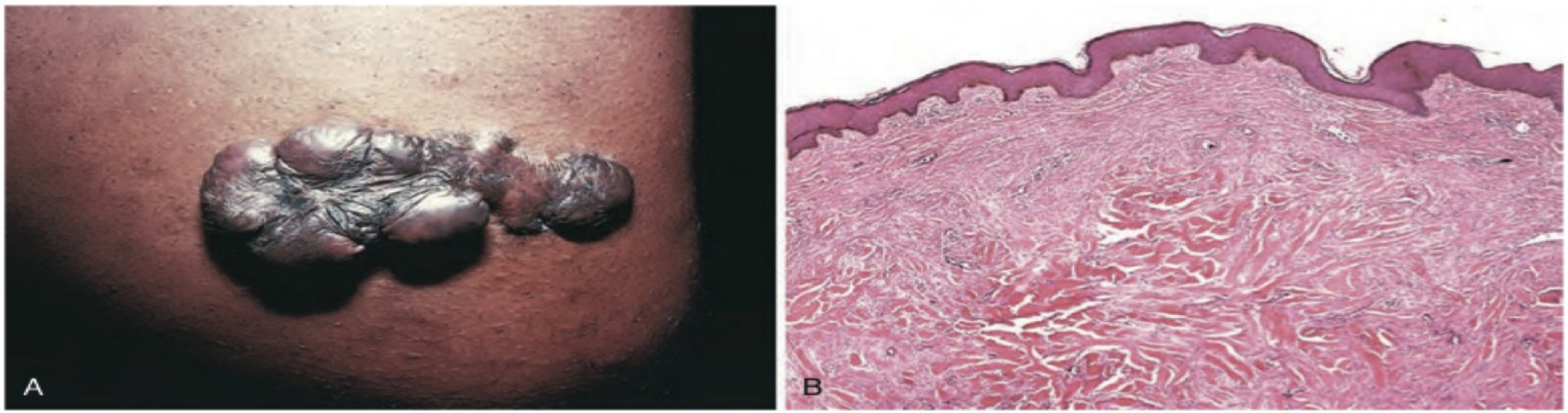


Figure 3-32 Keloid. **A,** Excess collagen deposition in the skin forming a raised scar known as *keloid*. **B,** Note the thick connective tissue deposition in the dermis. (**A,** From Murphy GF, Herzberg AJ: Atlas of Dermatopathology. Philadelphia, WB Saunders, 1996, p 219; **B,** Courtesy Z. Argenyi, MD, University of Washington, Seattle, Wash.)

C-Exuberant **granulation** tissues



4 Others

- Hernia
- Cyst

Factors affecting wound healing

Systemic factors

- Age
- Nutrition
- Trauma
- Metabolic diseases
- Immunosuppression
- Connective tissue disorders
- smoking

Local factors

- Mechanical injury
- Infection
- Edema
- Topical agents
- Ionizing radiation
- Necrotic tissue
- Low oxygen tension
- Foreign bodies